

Global Superstore

SUBJECT:DATA VISUALIZATION

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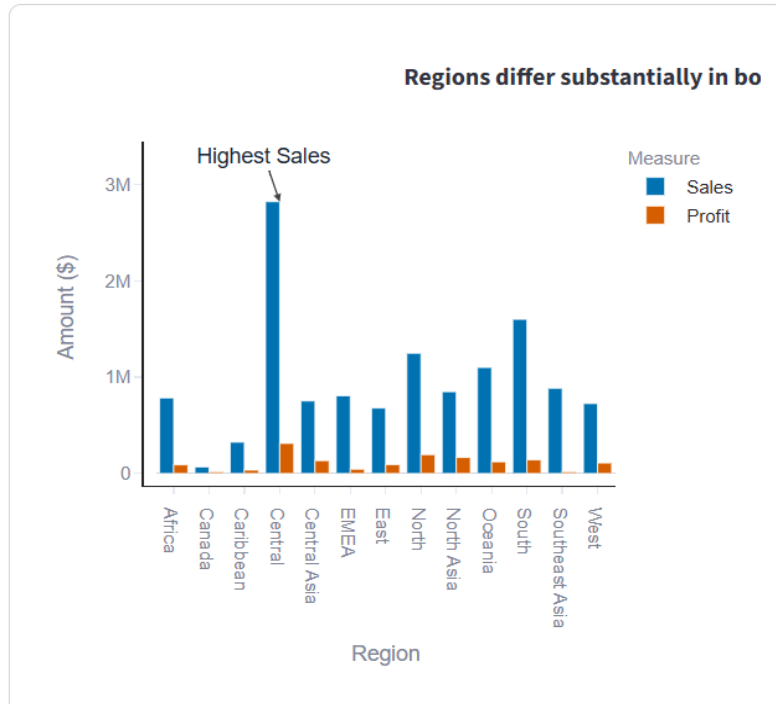
Introduction

- The **Global Superstore Data Visualization Project** is a business analytics project that aims to transform raw sales data into meaningful insights using **Python, Plotly, and Streamlit**. The Global Superstore dataset contains detailed information about customer orders, including sales, profit, discounts, shipping details, product categories, customer segments, and geographical locations across multiple international markets.
- The project uses **Python** for data preprocessing and analysis, **Plotly** for creating interactive and visually appealing charts, and **Streamlit** for developing an interactive dashboard that enables users to explore business insights through filters and dynamic visualizations. The dashboard presents key performance indicators (KPIs), trend analysis, geographical analysis, profitability analysis, and product performance in a clear and user-friendly format.

Dataset: <https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Data Visualization:

1. Which regions generate the highest sales and profit?

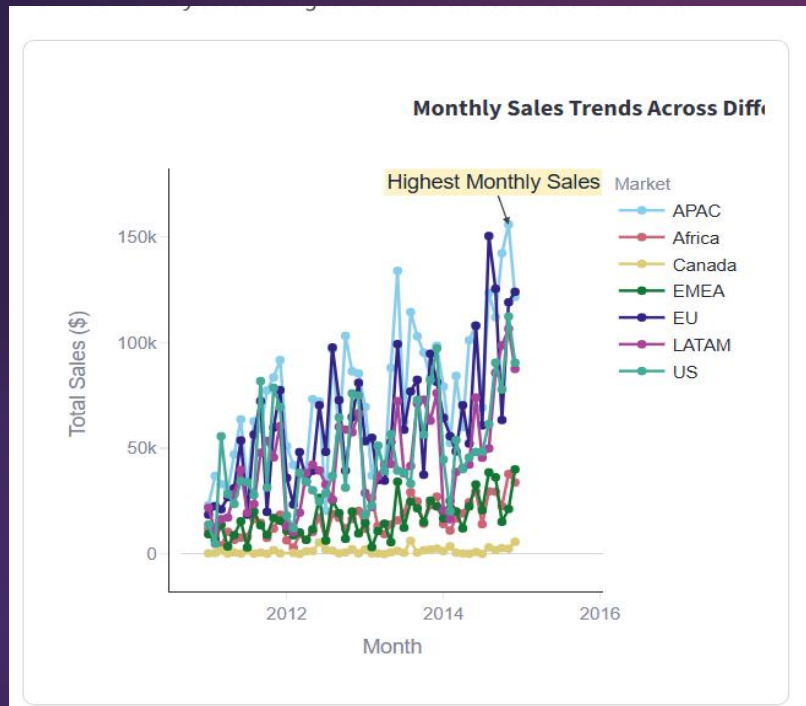


Business insight:

Central generates the highest total sales (\$2,822,303).

Central generates the highest total profit (\$311,404).

2. How has monthly sales changed over time across different markets?

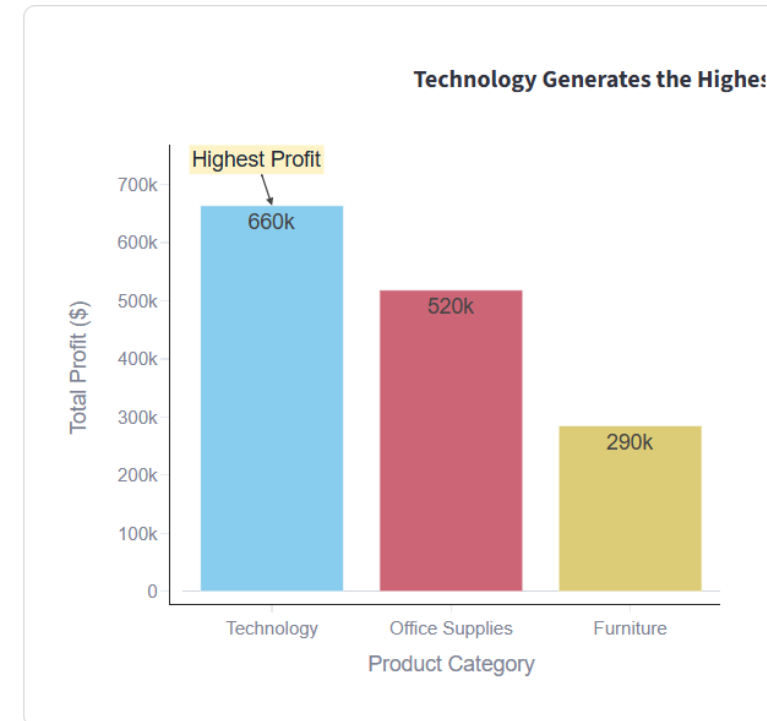


Business insight

The peak monthly market result is APAC in Nov 2014 (\$155,798).

The line chart preserves the notebook's month-by-market sales

3. WHICH PRODUCT CATEGORIES ARE MOST PROFITABLE?



Business insight

Technology is the most profitable category at \$663,779.

The category ranking is calculated from total profit, matching the notebook.

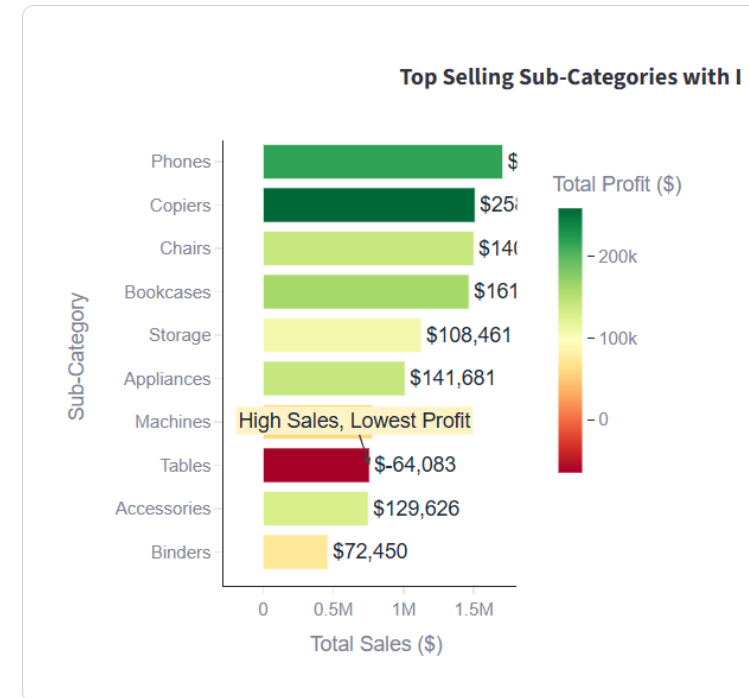
4. Does offering higher discounts actually increase profit?



Business insight

The highest-profit order earns \$8,400 at a 0% discount. The order-level scatter shows profit varies at each discount level rather than increasing uniformly with discount.

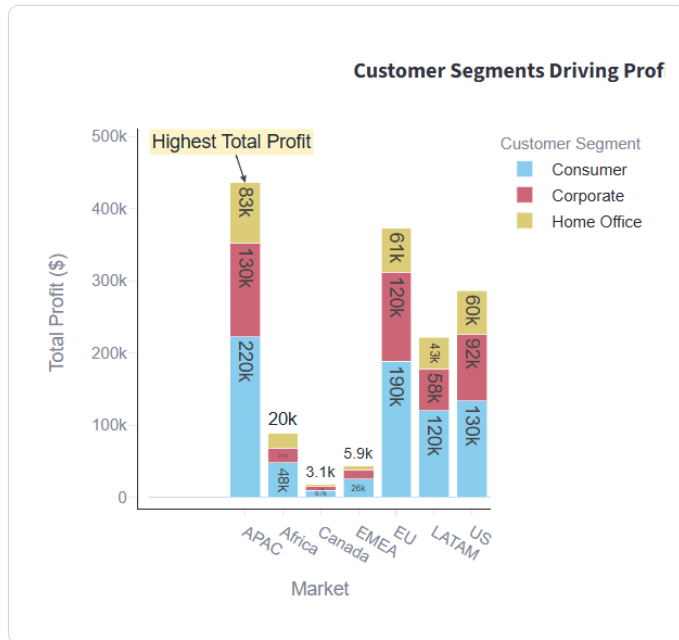
5. WHICH SUB-CATEGORIES GENERATE HIGH SALES BUT LOW PROFIT?



Business insight

Among the top 10 sub-categories by sales, Tables has the lowest profit (\$-64,083). The selection retains the notebook's sort: sales descending, then profit ascending, followed by the top 10.

6. Which customer segments are the most profitable across markets?

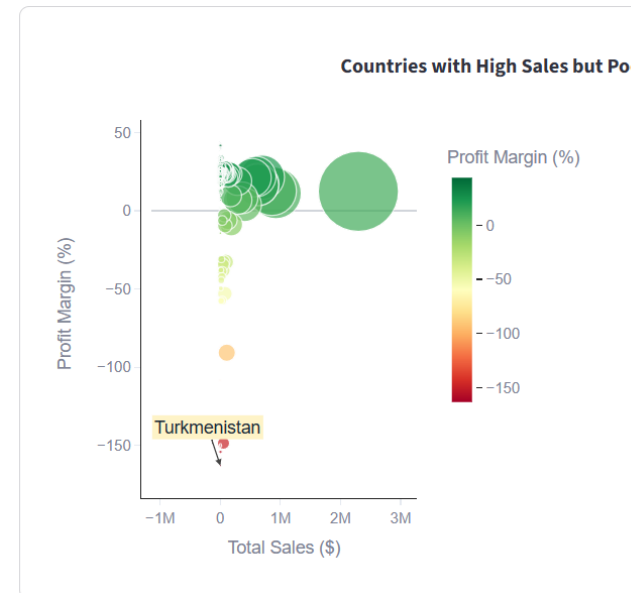


Business insight

APAC has the highest total market profit (\$436,000).

The largest market-segment profit contribution is Consumer in APAC (\$222,818).

7. Which countries have high sales but poor profit margins?

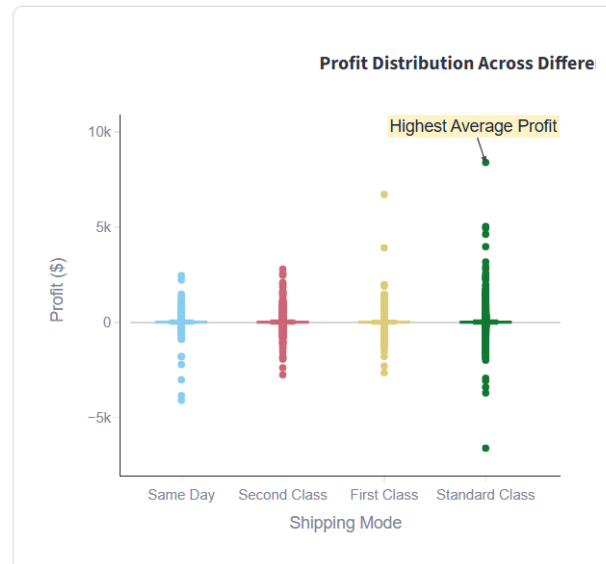


Business insight

Turkmenistan has the lowest profit margin (-162.93%) in the current selection.

Profit margin is retained as total profit divided by total sales, multiplied by 100.

8. How does shipping mode affect delivery performance and profit?

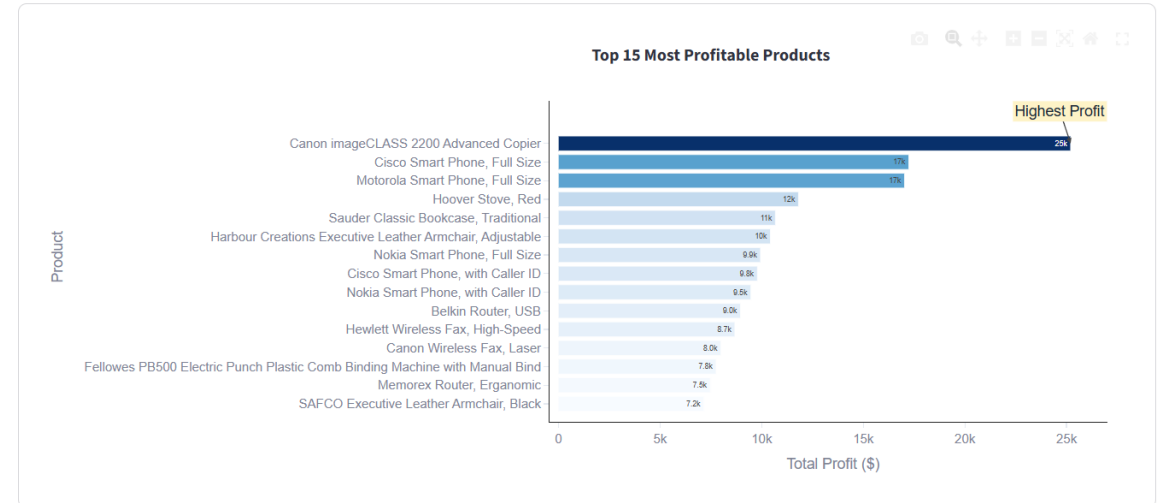


Business insight

Standard Class has the highest average profit (\$29) among shipping modes.

Delivery days are calculated as ship date minus order date; the notebook's visualization compares the resulting profit distributions by ship mode.

9. What are the Top 15 most profitable products?

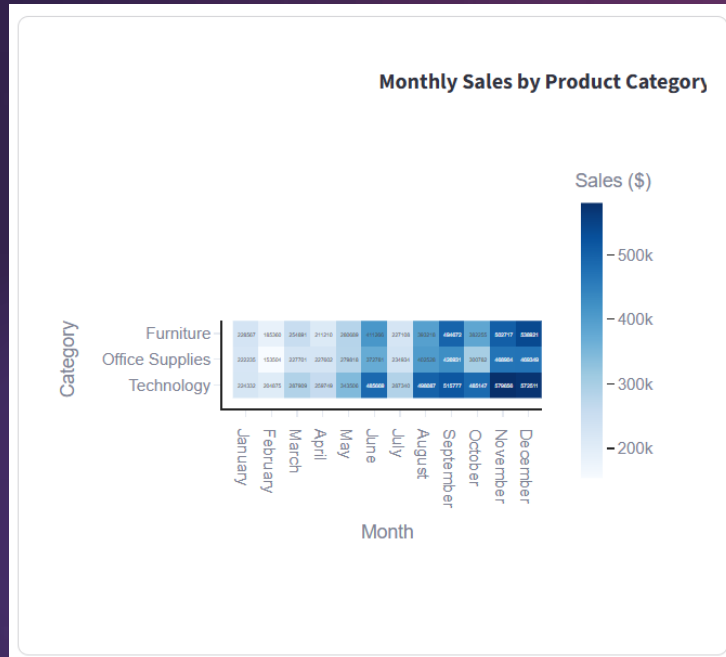


Business insight

Canon imageCLASS 2200 Advanced Copier is the most profitable product (\$25,200).

Products are ranked by summed profit and restricted to the top 15, exactly as in the notebook.

10. *How do sales and profit vary across months and categories?*



Business insight

The highest category-month sales value is Technology in November (\$579,656).
The highest category-month profit value is Technology in November (\$83,526).

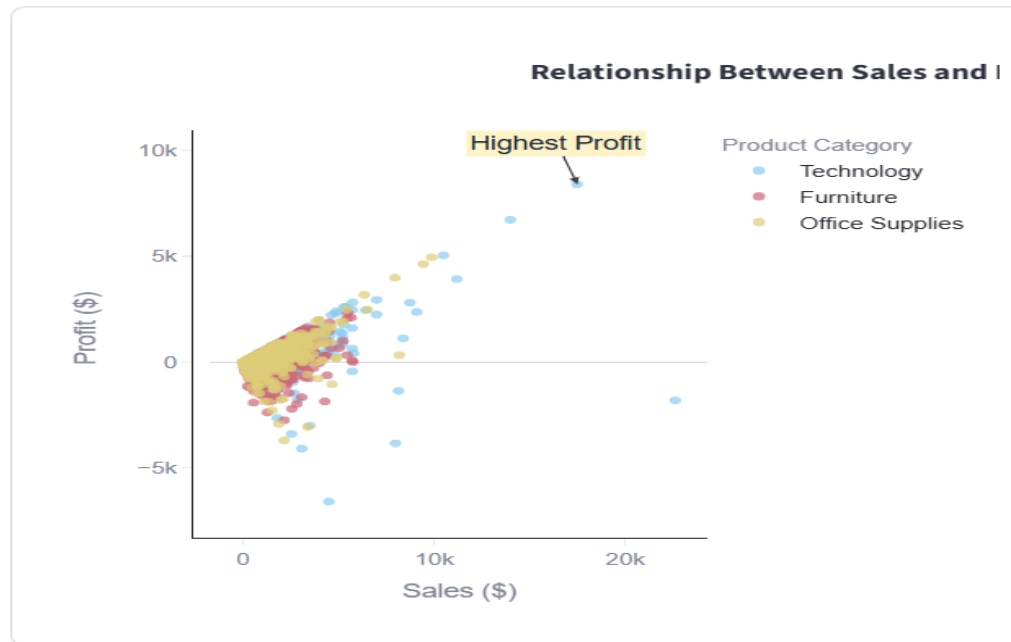
11. *Which cities consistently generate losses despite high sales?*



Business insight

Lagos has the lowest total profit among loss-making cities (\$-25,923).
The chart retains only negative-profit cities, sorts them by sales, and displays the top 15.

18. What is the relationship between sales and profit across all orders?



Business insight

The highest-profit order records 8,400 in profit on 17,500 in sales.

Each point represents an order line, coloured by product category as in the notebook.

Conclusion:

The analysis revealed that while overall sales are strong across several regions, higher sales do not always result in higher profits. Factors such as excessive discounts, low-margin products, and regional differences significantly influence profitability. Product categories like **Technology** emerged as major profit contributors, whereas some sub-categories and cities generated high sales but relatively low or negative profits. Customer segment analysis highlighted the segments that contribute most to revenue and profit, while shipping mode analysis showed the relationship between delivery methods and profitability. Seasonal trend analysis also identified periods of peak sales and profit, enabling better forecasting and business planning.

The interactive Streamlit dashboard enables users to explore these insights dynamically through filters and visualizations, making it easier for decision-makers to monitor business performance and identify areas for improvement. Overall, this project demonstrates the value of data visualization in supporting strategic business decisions and improving operational performance.

Links:-

Github link for plotly chart: <https://github.com/Naiya23/dataviz-exercises-naiyabenpatel/blob/main/dv-final.ipynb>

Streamlit link: <https://jfdgsf.streamlit.app/>

Thank You!